DBMS TASK 5

Building the OnlineShoppingDB Schema

Purpose

Create the database and define the tables, keys, and relationships needed to support customer orders, product line items, and payments.

A. Create Database and Tables

```
CREATE DATABASE IF NOT EXISTS OnlineShoppingDB;
USE OnlineShoppingDB;

CREATE TABLE Customers (
    CustomerID INT AUTO_INCREMENT PRIMARY KEY,
    CustomerName VARCHAR(100) NOT NULL,
    Email VARCHAR(120) NOT NULL UNIQUE,
    Phone VARCHAR(20),
    Address VARCHAR(255)
);

CREATE TABLE Products (
    ProductID INT AUTO_INCREMENT PRIMARY KEY,
    ProductName VARCHAR(120) NOT NULL,
    Category VARCHAR(60) NOT NULL,
    Price DECIMAL(10,2) NOT NULL,
    Stock INT NOT NULL DEFAULT 0,
    CONSTRAINT chk_product_price CHECK (Price >= 0),
    CONSTRAINT chk_product_stock CHECK (Stock >= 0)
);

CREATE TABLE Orders (
    OrderID INT AUTO_INCREMENT PRIMARY KEY,
    CustomerID INT NOT NULL,
    OrderDate DATE NOT NULL,
    Status VARCHAR(20) NOT NULL DEFAULT 'Pending',
    CONSTRAINT fk_orders_customer
        FOREIGN KEY (CustomerID) REFERENCES Customers(CustomerID),
    CONSTRAINT chk_order_status
        CHECK (Status IN ('Pending','Processing','Shipped','Delivered','Cancelled'))
);

CREATE TABLE OrderDetails (
    OrderDetailID INT AUTO_INCREMENT PRIMARY KEY,
    OrderID INT NOT NULL,
    ProductID INT NOT NULL,
    Quantity INT NOT NULL,
    UnitPrice DECIMAL(10,2) NOT NULL,
    CONSTRAINT fk_details_order
        FOREIGN KEY (OrderID) REFERENCES Orders(OrderID),
    CONSTRAINT fk_details_product
        FOREIGN KEY (ProductID) REFERENCES Products(ProductID),
    CONSTRAINT chk_quantity CHECK (Quantity > 0),
    CONSTRAINT chk_unit_price CHECK (UnitPrice >= 0)
);

CREATE TABLE Payments (
    PaymentID INT AUTO_INCREMENT PRIMARY KEY,
```

```
    OrderID INT NOT NULL,
    PaymentMethod VARCHAR(30) NOT NULL,
    Amount DECIMAL(10,2) NOT NULL,
    PaymentStatus VARCHAR(20) NOT NULL DEFAULT 'Pending',
    CONSTRAINT fk_payments_order
        FOREIGN KEY (OrderID) REFERENCES Orders(OrderID),
    CONSTRAINT chk_payment_amount CHECK (Amount >= 0),
    CONSTRAINT chk_payment_status
        CHECK (PaymentStatus IN ('Pending','Paid','Failed','Refunded'))
);
```

B. Confirm the Schema

```
USE OnlineShoppingDB;
SHOW TABLES;

DESCRIBE Customers;
DESCRIBE Products;
DESCRIBE Orders;
DESCRIBE OrderDetails;
DESCRIBE Payments;
```

Expected check: five tables should be listed. Run the commands in MySQL Workbench and insert screenshots showing the real output.

C. Relationship Summary

Relationship	Meaning
Customers → Orders	Each customer may place several orders.
Orders → OrderDetails	An order can contain multiple products.
Products → OrderDetails	A product may appear on multiple order lines.
Orders → Payments	Payment records are associated with an order.

Closing Remarks

The schema creates the core entities and uses primary and foreign keys to define how customers, products, orders, order lines, and payments are connected.

SQL CODE — MYSQL WORKBENCH

Execute first. If tables already exist, do not rerun CREATE TABLE statements unless you intentionally reset the database.

```
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USE OnlineShoppingDB;

CREATE TABLE Customers (
 CustomerID INT AUTO_INCREMENT PRIMARY KEY,
 CustomerName VARCHAR(100) NOT NULL,
 Email VARCHAR(120) NOT NULL UNIQUE,
 Phone VARCHAR(20), Address VARCHAR(255)
);

CREATE TABLE Products (
 ProductID INT AUTO_INCREMENT PRIMARY KEY,
 ProductName VARCHAR(120) NOT NULL,
 Category VARCHAR(60) NOT NULL,
 Price DECIMAL(10,2) NOT NULL CHECK (Price >= 0),
 Stock INT NOT NULL DEFAULT 0 CHECK (Stock >= 0)
);

CREATE TABLE Orders (
 OrderID INT AUTO_INCREMENT PRIMARY KEY,
 CustomerID INT NOT NULL,
 OrderDate DATE NOT NULL,
 Status VARCHAR(20) NOT NULL DEFAULT 'Pending',
 FOREIGN KEY (CustomerID) REFERENCES Customers(CustomerID),
 CHECK (Status IN ('Pending','Processing','Shipped','Delivered','Cancelled'))
);

CREATE TABLE OrderDetails (
 OrderDetailID INT AUTO_INCREMENT PRIMARY KEY,
 OrderID INT NOT NULL,
 ProductID INT NOT NULL,
 Quantity INT NOT NULL CHECK (Quantity > 0),
 UnitPrice DECIMAL(10,2) NOT NULL CHECK (UnitPrice >= 0),
 FOREIGN KEY (OrderID) REFERENCES Orders(OrderID),
 FOREIGN KEY (ProductID) REFERENCES Products(ProductID)
);

CREATE TABLE Payments (
 PaymentID INT AUTO_INCREMENT PRIMARY KEY,
 OrderID INT NOT NULL,
 PaymentMethod VARCHAR(30) NOT NULL,
 Amount DECIMAL(10,2) NOT NULL CHECK (Amount >= 0),
 PaymentStatus VARCHAR(20) NOT NULL DEFAULT 'Pending',
 FOREIGN KEY (OrderID) REFERENCES Orders(OrderID),
 CHECK (PaymentStatus IN ('Pending','Paid','Failed','Refunded'))
);

SHOW TABLES;
DESCRIBE Customers;
DESCRIBE Products;
DESCRIBE Orders;
DESCRIBE OrderDetails;
DESCRIBE Payments;
```